Pickup And Delivery With Deadline Using Drones In Obstructed Course

Submitted as a partial fulfillment of Bachelor of Technology in Computer Science & Engineering
of
Maulana Abul Kalam Azad University of Technology
(Formerly known as West Bengal University of Technology)

Project Report

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ABSTRACT

The rapid evolution of smart logistics, emergency medical dispatch, and decentralized supply chains has accelerated the integration of Unmanned Aerial Vehicles (UAVs) into autonomous transport networks. Conventional drone logistics frameworks are primarily optimized for centralized warehouse-to-customer topologies. While effective for routine commercial distribution, these rigid hub-and-spoke models lack the dynamic operational agility required for point-to-point routing between arbitrary coordinates—an essential capability in time-critical emergency scenarios, such as the transport of pharmaceuticals, biological specimens, and emergency payloads.

This report proposes a novel end-to-end framework, the Hybrid Zone-Based Multi-Token Passing (HZ-MTP) system, engineered to manage variable-pickup to variable-destination delivery requests across highly obstructed urban environments. Operating on complex 256×256 and 1024×1024 occupancy grids representing realistic urban layouts, the architecture structures high-fidelity obstacles into geometric convex hulls to streamline path-planning computations. Task scheduling and resource allocation are governed by the Zone-Based Multi-Token Task Assignment with Deadline-Proximity (Z-MTD) algorithm. This scheduling mechanism dynamically monitors drone state spaces (availability, location, and battery state-of-charge) and utilizes localized zone tokens containing shared path reservation tables. To guarantee strict delivery windows, tasks are assigned via a dual-criteria heuristic that prioritizes earliest delivery deadlines alongside physical proximity to minimize transit overhead.

Complementing this task allocation layer, the report introduces the Perimeter Extraction, Recursive Subdivision, and Smoothing (PRS) algorithm. Rather than executing computationally expensive global grid searches (such as conventional A* or Dijkstra), the routing engine constructs a direct line-of-sight vector from source to destination. By isolating only the intersecting obstacle perimeters, the algorithm applies the Moore-Neighbor Tracing technique to extract the obstacle's continuous boundary. This boundary is subsequently expanded via Minkowski-sum inflation using a safety margin ($\varepsilon = 0.05$ m) to mitigate physical collisions and localization errors. Obstacle bypass trajectories are evaluated through local perimeter-slicing and a recursive waypoint-pruning routine (smoothPath), minimizing trajectory length, sharp-turn kinematics, and aerodynamic drag.

The integrated framework was validated through extensive multi-agent simulations on high-resolution urban models. The experimental findings demonstrate that the HZ-MTP architecture significantly minimizes path-planning computational latency, reduces routing distance, and scales efficiently relative to traditional A*-guided multi-agent systems. Crucially, the localized token-passing paradigm successfully resolves spatial path conflicts and eliminates deadlock configurations, ensuring high deadline satisfaction rates within dense, dynamic urban airspaces.

Keywords: *Autonomous UAV Logistics, Multi-Agent Task Assignment, Convex Hull Obstacle Avoidance, Minkowski-Sum Inflation, Perimeter-Slicing Routing, Token-Passing Orchestration.*

Chapter 1 : Introduction

1.1 Background

The increasing demand for rapid package delivery has encouraged researchers and industries to adopt autonomous technologies for logistics. Drone-based delivery systems have emerged as an efficient solution for last-mile delivery due to their high mobility, reduced travel time, and ability to bypass road congestion. Companies such as Amazon, UPS, DHL, and Zipline have demonstrated the practical potential of drone-assisted logistics.

Although drones offer numerous advantages, safe navigation in urban environments remains a challenging problem. A city contains numerous high-rise buildings, restricted zones, and other obstacles that prevent a drone from traveling in a straight line. Therefore, an efficient path planning algorithm is essential for ensuring safe, collision-free, and time-efficient deliveries.

Traditional path planning algorithms such as Dijkstra's Algorithm and A* Search guarantee shortest or near-shortest paths but often require exploring a large number of nodes in high-resolution maps. As map size increases from 256 × 256 to 1024 × 1024 grids, computational cost also increases significantly. Consequently, there is a need for a routing algorithm that minimizes unnecessary search while maintaining path quality.

Conventional e-commerce delivery systems generally operate on a warehouse-to-customer model, where products are stored in centralized warehouses and transported to customers through a multi-stage logistics network. While this approach is effective for large-scale commerce, it may not be suitable for urgent deliveries that require immediate transportation between arbitrary locations. The system proposed in this project supports variable pickup and variable destination delivery, allowing a drone to collect an item from any registered pickup point and deliver it directly to any specified destination. Because the transportation is performed by autonomous drones rather than ground vehicles, the system can avoid road congestion and significantly reduce delivery time. Such a framework is highly beneficial in emergency scenarios, including the delivery of medicines, blood samples, medical equipment, legal documents, and other critical items where rapid response is essential.

1.2 Objectives of the Project

The primary objective of this project is to design and develop an intelligent Drone-Based Delivery System integrated with a novel shortest path algorithm for efficient package transportation in urban environments within deadlines. The specific objectives are as follows:

- To develop a drone-based delivery framework that supports variable pickup to variable destination delivery instead of the conventional warehouse-to-customer delivery model.
- To convert building obstacles into hulls and extract their hull vertices to simplify obstacle representation and improve path planning efficiency.
- To design and implement a novel shortest path algorithm that computes collision-free routes by processing only the obstacles intersecting the source-to-destination path.
- To minimize travel distance and routing computation time by avoiding unnecessary exploration of the entire search space.
- To generate delivery tasks consisting of pickup location, delivery location, and delivery deadline for efficient delivery management.

- To develop an intelligent multi-drone task assignment mechanism that allocates tasks based on geographical proximity and deadline feasibility.
- *To compare the proposed shortest path algorithm with the A algorithm** in terms of path length, computation time, node exploration, and overall routing efficiency.
- To evaluate the performance of single-token and multi-token routing strategies and analyze their impact on scalability and execution time.
- To provide an efficient solution for emergency logistics, such as the delivery of medicines, blood samples, emergency equipment, and important documents, where rapid point-to-point transportation is critical.

1.3 Scope of the Work

The scope of the proposed project includes the following:

- Develop an autonomous drone-based delivery system for efficient package transportation in urban environments.
- Support variable pickup to variable destination delivery, unlike conventional warehouse-to-customer logistics systems.
- Model real-world city environments using 256 × 256 and 1024 × 1024 occupancy grid maps.
- Represent multi-storey buildings as hulls to simplify obstacle modeling and improve routing efficiency.
- Design and implement a novel hull-based shortest path algorithm for collision-free navigation.
- Compute the shortest feasible path by considering only the obstacles intersecting the direct source-to-destination path, thereby reducing unnecessary search operations.
- Support multiple drones operating simultaneously for handling multiple delivery requests.
- Generate delivery tasks containing pickup location, delivery location, and delivery deadline.
- Perform intelligent task assignment based on drone proximity, zone allocation, and deadline feasibility.
- Simulate the complete delivery workflow, including task generation, drone assignment, route planning, pickup, delivery, and task completion.
- Compare the performance of the proposed shortest path algorithm with the A* algorithm using metrics such as path length, computation time, routing efficiency, and obstacle handling.
- Analyse and compare single-token and multi-token routing approaches to evaluate computational performance and scalability.
- Demonstrate the applicability of the proposed system for emergency delivery services, including medicines, blood samples, medical equipment, and urgent documents, where rapid point-to-point transportation is essential.

Chapter 2: Literature Review

2.1 Introduction to Drone Logistics and Task Allocation

The integration of Autonomous Aerial Vehicles (UAVs or drones) into logistics has gained significant traction due to the rapid growth of e-commerce, the development of smart cities, and the demand for rapid transport of critical cargo. Unlike traditional road transportation, which is constrained by traffic congestion and infrastructure layouts, drones offer the advantage of direct, line-of-sight flight paths, yielding substantial reductions in delivery times.

Beyond commercial applications, drone delivery plays a critical role in emergency logistics, such as the rapid transport of:

- Medical supplies, vaccines, and blood samples between healthcare facilities.
- Emergency equipment and diagnostic kits to remote disaster areas.
- Time-sensitive medical documents and organ transplants.

Most commercial drone systems currently follow a **warehouse-to-customer** logistics model, where drones originate from a single centralized hub and fly to predefined customer destinations. While structured and easier to manage, this model fails to support applications that require peer-to-peer or point-to-point transportation. For emergency operations, a more flexible **variable pickup to variable destination** delivery framework is required. This framework must handle dynamic task arrival, satisfy strict delivery deadlines, and plan collision-free trajectories in obstructed urban spaces.[2]

2.2 Multi-Agent Path Finding (MAPF) and Multi-Agent Pickup & Delivery (MAPD)

To coordinate fleets of autonomous agents, researchers have extensively studied **Multi-Agent Path Finding (MAPF)**. The goal of MAPF is to find collision-free paths for a team of agents from their respective start positions to their target destinations while optimizing objectives such as flowtime or make span. While MAPF establishes the theoretical foundation for collision avoidance and path coordination, it operates under the assumption of static environments with predetermined, one-time start and goal coordinates.

To address continuous operations where tasks arrive dynamically, the **Multi-Agent Pickup and Delivery (MAPD)** framework was introduced. In MAPD, agents continuously execute tasks, each consisting of a pickup location and a delivery location.[3]

2.2.1 Deadline-Aware MAPD (MAPD-D)

In emergency logistics, tasks are associated with strict delivery deadlines. This has led to the development of deadline-aware scheduling models, such as the **Deadline-Aware Multi-Agent**

Tour Planning (DA-MATP) problem. To resolve these challenges, algorithms like **ROSETTA** combine Prioritized Planning with Large Neighbourhood Search (LNS) to maximize the percentage of tasks delivered before their deadlines. Further variations introduce dynamic task swapping, task switching, and token-passing heuristics to minimize total task tardiness.[3]

2.3 Path Planning Algorithms and Spatial Representations

The core of any multi-agent routing system is the single-agent shortest path algorithm.

2.3.1 Graph Search Algorithms (Dijkstra and A*)

Traditional path planning relies on grid-based graph search, primarily **Dijkstra's algorithm** and **A***.

- **Dijkstra's algorithm** guarantees mathematical optimality by expanding search nodes uniformly in all directions. However, this exhaustive search is computationally prohibitive for real-time operations on large maps.
- *The **A algorithm*** incorporates a heuristic function (e.g., Euclidean or Manhattan distance) to direct the search toward the goal, expanding fewer nodes. However, **A*** still faces scalability challenges. In dense urban environments with complex building structures, the number of expanded grid nodes explodes as the grid resolution increases, making real-time re-planning difficult.

2.3.2 Spatial and Obstacle Representations

How obstacles are represented in memory significantly impacts path-planning efficiency:

1. **Occupancy Grids**: The environment is discretized into a uniform grid of cells, each marked as free or occupied. While simple, grid representation forces pathfinders to evaluate individual grid cells, creating large search spaces and causing "stair-step" kinematic paths.
2. **Geometric Representations**: Methods such as **Visibility Graphs** and **Polygons** simplify the environment by modeling obstacles geometrically. By representing multi-storey buildings as **Hulls**, algorithms can ignore the internal space of buildings and focus only on the boundary vertices. This significantly reduces the size of the search graph and allows continuous-vector path planning.[4]

2.4 Limitations of Conventional Single Token Passing (TP)

In conventional MAPD systems, **Token Passing (TP)** is a standard decentralized method for agent coordination. In this scheme, a single global data structure—the **Token**—is maintained. The token

stores the current task assignments, path reservations, and status of all agents. While the single-token protocol mathematically guarantees collision-free routing by allowing only one agent to update its path at any given time, it exhibits severe operational bottlenecks when scaled to real-world drone delivery fleets:

2.4.1. Sequential Planning Bottleneck

The primary operational constraint of Single Token Passing is that **path planning is strictly sequential**. Only the single agent that currently holds the token is permitted to plan its path and write reservations. All other agents requiring path modifications or task assignments must wait in a queue. As the fleet size grows, the cumulative waiting time in the token queue increases linearly, leading to significant coordination delays.

2.4.2. Poor Fleet Scalability

The sequential nature of the token loop makes it highly unscalable. While the approach functions efficiently for small fleets (e.g., 5 to 10 warehouse robots), it degrades rapidly when applied to hundreds of drones managing thousands of dynamic delivery requests. High token contention causes the system's throughput to plateau, rendering it unusable for city-scale logistics.

2.4.3. High Communication Overhead

Under the single-token protocol, every drone in the fleet must communicate with the same centralized token server. As the geographical area expands, drones located kilometers away must continuously transmit path requests and updates to the centralized server. This central communication hub creates network congestion and high bandwidth usage, regardless of whether the drones are operating in close proximity or on completely opposite sides of a city.

2.4.4. Increased Delivery Latency

When urgent tasks (such as critical medical deliveries) enter the system, they cannot be prioritized or assigned immediately. A free drone must wait for its turn to acquire the global token before it can assign the urgent task to itself and plan a path. This inability to bypass the sequential queue increases task tardiness and compromises deadline satisfaction for time-critical payloads.

2.4.5. Low Drone Utilization and Battery Wastage

Because drones must wait for token access before initiating flight or updating paths, they spend significant periods idling on charging pads or hovering in the air. For battery-powered UAVs, hovering while waiting for a path update directly translates to wasted battery capacity. This reduces the effective operational range and flight time of the drones.

2.4.6. Centralized Single Point of Failure

The Single Token Passing architecture depends entirely on the availability of the centralized token server. If the server experiences a network dropout, hardware failure, or software crash, the token becomes inaccessible. Because drones cannot plan paths without the token, the entire fleet is immediately grounded, creating a major vulnerability in critical logistics networks.

2.4.7. Global Search Space Overhead

Path planning within a single-token system is performed over the entire global map. For city-wide routing, this requires looking up obstacles across the entire metropolitan area. The resulting search space is unnecessarily large, which increases the computational time for each planning step and limits the system's ability to recalculate paths in real time.[2]

2.5 Transition to Hybrid Zone-Based Multi-Token Passing

To address these limitations, modern drone routing architectures employ **spatial partitioning**. By dividing the metropolitan area into distinct, independent geographic zones, the single global token can be replaced with multiple localized tokens.

In a **Hybrid Zone-Based Multi-Token Passing** model:

- Each zone maintains its own local task list, active drone fleet, and zone-specific token.
- Path planning and task allocation are executed in parallel across all zones, eliminating the sequential bottleneck.
- Communication is localized, reducing global network overhead.
- A failure in a single zone is isolated, allowing the rest of the city's drone network to continue operating normally.

Chapter 3 : Problem Description

This work addresses the Drone-Based Multi-Agent Pickup and Delivery with Deadlines (D-MAPD-D) problem in a real-life urban environment. The objective is to coordinate a fleet of autonomous drones to perform dynamic pickup and delivery operations efficiently while satisfying delivery deadlines, minimizing travel distance, and maximizing the overall performance of the delivery system. As the number of delivery requests and drones increases, efficient coordination becomes increasingly challenging due to dynamic task arrivals, concurrent drone operations, obstacle avoidance, shortest path computation, and real-time task allocation.[3]

The operational environment is represented using a city occupancy grid map extracted from benchmark map files. The city is modelled as a connected graph

$$G = (V, E) \tag{1}$$

where Vdenotes the set of traversable locations extracted from the occupancy grid map and Erepresents the feasible movements between adjacent traversable locations. Static obstacles such as buildings and restricted regions are identified from the map and represented by their hull boundaries. These hulls define non-traversable regions during navigation. Since the height of every obstacle is assumed to be greater than the maximum permissible flight altitude of the drones, drones are required to navigate around obstacles rather than flying over them. Pickup locations, delivery locations, and charging stations are selected from the traversable nodes of the graph.
A fleet of autonomous drones

$$D = \{d_1, d_2, d_3, \dots, d_m\} \tag{2}$$

operates over the city environment. At every discrete time step t, drone $d_i$occupies a traversable node

$$l_i(t) \in V \tag{3}$$

and may either hover or move to an adjacent traversable node. The system continuously receives dynamically generated delivery requests represented by the task set

$$T = \{\tau_1, \tau_2, \dots, \tau_n\} \tag{4}$$

where each task

$$\tau_j = (v_j^p, v_j^d, d_j) \tag{5}$$

consists of a pickup location v_j^p, a delivery location v_j^d, and a soft delivery deadline d_j. A task is considered completed when a drone reaches the pickup location, collects the package, and subsequently delivers it to the designated destination. The completion time of task $\tau_j$is denoted by c_j, where one unit of travel distance corresponds to one unit of execution time.[1]
As delivery requests arrive continuously, the system must determine which drone should execute each task while simultaneously computing a collision-free route for that drone. In a large-scale urban environment, conventional centralized scheduling approaches suffer from communication bottlenecks and poor scalability because all drones depend on a single task allocation mechanism. Furthermore, assigning tasks without considering the geographical distribution of delivery requests increases travel distance, delivery delay, and drone idle time. Similarly, generating only a single feasible route does not guarantee minimum travel distance, as multiple obstacle-free paths may exist between the source and destination.

To improve scalability, the operational area is partitioned into multiple geographical zones

$$Z = \{Z_1, Z_2, \dots, Z_k\} \tag{6}$$

where each zone contains a subset of drones responsible for serving delivery requests generated within that region. Task allocation is therefore performed independently inside each zone, enabling multiple delivery requests to be processed simultaneously while reducing scheduling latency and communication overhead. Within each zone, delivery tasks must be assigned to the most suitable available drone such that task waiting time and delivery delay are minimized while maintaining balanced utilization of the drone fleet.

After task assignment, every drone must determine the shortest collision-free route from its current location to the pickup location and subsequently to the delivery destination. Let

$$\mathcal{P} = \{P_1, P_2, \dots, P_m\} \tag{7}$$

represent the set of all feasible obstacle-free paths between two locations. Each feasible path satisfies

$$P_i \cap O = \emptyset, \tag{8}$$

where O denotes the set of hull obstacles. The total length of each feasible path is given by

$$L(P_i) = \sum_{k=1}^{r-1} d(v_k, v_{k+1}), \tag{9}$$

where $d(v_k, v_{k+1})$denotes the Euclidean distance between two consecutive navigation points. The optimal path is defined as

$$P^* = \arg\min_{P_i \in \mathcal{P}} L(P_i), \tag{10}$$

which represents the shortest feasible collision-free route between the source and destination.

The delivery tardiness of task $\tau_j$is defined as

$$\epsilon_j = \max(0, c_j - d_j), \tag{11}$$

where

$$\epsilon_j = 0 \tag{12}$$

indicates that the task is completed within its deadline, whereas

$$\epsilon_j > 0 \tag{13}$$

indicates that the delivery deadline has been exceeded.

Therefore, the D-MAPD-D problem consists of simultaneously determining (i) the optimal assignment of dynamically arriving delivery tasks to multiple drones, (ii) the shortest feasible collision-free path for every assigned drone, and (iii) an efficient coordination strategy that allows multiple drones to operate concurrently across different city zones while satisfying delivery deadlines. The overall objective is to minimize delivery tardiness , total travel distance and to maximize the no of feasible task within deadline while maximizing system throughput, reducing task allocation latency, improving drone utilization, and providing a scalable multi-agent drone delivery framework capable of operating efficiently over large-scale city maps under continuously changing delivery demands.[2]

Chapter 4 : Proposed Solution

To address the limitations of centralized and computationally heavy logistics networks, we propose an integrated architectural framework founded on two main operational pillars: Dynamic Fleet Task Allocation and Spatial Path Routing and Validation.

The first pillar manages the intelligent distribution of multi-destination delivery workloads across the decentralized drone network using the proposed **Zone-Based Multi-Token Task Assignment with Deadline-Proximity (Z-MTD)** algorithm. This mechanism isolates tasks into localized geographic zones and uses independent scheduling tokens to simultaneously allocate tasks based on immediate deadline constraints and drone availability.[2]

The second pillar resolves spatial navigation challenges through the proposed **Perimeter Extraction, Recursive Subdivision, and Smoothing (PRS)** algorithm. Once a task is matched to a specific agent, this routing engine constructs direct line-of-sight vectors, maps obstacle obstructions as continuous geometric boundaries, and executes local perimeter-slicing and recursive waypoint pruning to generate collision-free, aerodynamic trajectories. By combining task assignment and path planning into a single, cohesive framework, the proposed solution guarantees real-time multi-drone coordination and high deadline satisfaction rates within highly obstructed urban airspaces.

4.1 Hybrid Zone-Based Multi-Token Model

To overcome the performance bottlenecks, high computational overhead, and single-point-of-failure vulnerabilities of centralized single-token architectures, we propose a **Hybrid Zone-Based Multi-Token Passing (HZ-MTP)** model. By geographically partitioning the global operational airspace Ω into a set of K disjoint zones $\{\Omega_1, \Omega_2, \dots, \Omega_K\}$, the scheduling and route reservation spaces are completely decentralized. Each zone operates as an independent scheduling domain, enabling high-concurrency, localized task assignment and route planning.

The HZ-MTP system model comprises three core components in each zone Ω_z:

- **Local Drone Fleet ($\mathcal{A}_z$):** A dynamically updated subset of drones physically located within the boundaries of zone z.
- **Local Task Set ($\mathcal{T}_z$):** A dynamic registry tracking all active, unassigned pickup-and-delivery tasks whose pickup coordinates originate within the geographical boundaries of zone z.
- **Independent Zone Token (**token_z**):** A decentralized local data structure containing:
    - **The Local Task Table:** A synchronized list of active, pending, and in-progress tasks.

- **The Spatial Path Reservation Table ($\mathcal{P}_z$):** A continuous spacetime map storing the collision-free trajectories of all active drones operating within the zone to prevent inter-agent collisions.

Since every token operates independently, multiple zones can perform task allocation and route reservation simultaneously without global synchronization locks, drastically reducing computational latency and scheduling overhead.

Fig 1 : Zone Partition System



4.1.1 Mathematical Formulation & Execution Logic of Task Assignment

The task assignment process is executed dynamically in a loop, utilizing the localized zone token to safely allocate tasks to available drones. The execution sequence is structured into three continuous, cyclical phases:

Phase 1: Local Token Acquisition and Fleet Registration

At each timestep t, every drone $a_i \in \mathcal{A}_z$ continuously monitors and registers its physical and operational state space:

$$\mathbf{x}_i = \langle \text{loc}(a_i), \text{batt}(a_i), \text{State}(a_i) \rangle \tag{14}$$

If a drone is unoccupied ($\text{State}(a_i) = \text{FREE}$), it actively issues a request to acquire the local token_z. The token acts as a mutual exclusion lock; only the drone holding token_z is permitted to read the local task list $\mathcal{T}_z$ and write its planned trajectory to the reservation table $\mathcal{P}_z$. This mechanism guarantees that no two drones can be assigned the same task or generate conflicting routes.[1]

Phase 2: Urgency-Based Initial Task Allocation

Upon acquiring token_z, the idle agent a_i identifies the set of unassigned tasks $\mathcal{T'}_z \subseteq \mathcal{T}_z$. To maximize overall system performance and avoid deadline violations, the system applies an **Urgency-First Allocation Rule**. The agent evaluates the delivery deadlines of all available tasks and selects the task τ_{j^*} with the most critical (earliest) absolute deadline:

$$\tau_{j^*} = \arg \min_{\tau_j \in \mathcal{T'}_z} \left(d_j^d \right) \tag{15}$$

Once $\boldsymbol{\tau_{j^*}}$ is selected:

1. The task is assigned to a_i, and the drone's status is updated: $\text{State}(a_i) \leftarrow \text{BUSY}$.
2. The task is removed from the active local task list $\mathcal{T}_z$.
3. A collision-free path $\mathcal{P}(a_i)$ is computed using the **Local Perimeter-Slicing Obstacle Avoidance (L-PSA)** algorithm from the current location to the pickup point $v_{j^*}^p$, and then to the delivery point $v_{j^*}^d$.
4. This path is written directly into token_z's reservation table ($\mathcal{P}_z$), after which the drone releases the token for other idle drones to acquire.[2]

Phase 3: Dynamic Proximity-Deadline Re-assignment

When an active drone completes its current delivery task at v_j^d, it instantly registers its updated status as available: $\text{State}(a_i) \leftarrow \text{FREE}$. Instead of remaining idle or returning to a central depot, the drone immediately requests token_z of its current zone z to claim a subsequent task.

To minimize unproductive, empty transit distance (the flight segment from the drone's current location to the next task's pickup point), the system switches from pure deadline urgency to a **Combined Deadline-Proximity Cost Metric** to evaluate the remaining tasks $\tau_j \in \mathcal{T}_z$:

$$\text{Cost}(\tau_j) = \alpha \cdot \text{dist}\left(\text{loc}(a_i), v_j^p\right) + \beta \cdot \left(d_j^d - t\right) \tag{16}$$

Where:

- $\text{dist}\left(\text{loc}(a_i), v_j^p\right)$ is the Euclidean distance from the drone's current position to the task's pickup location.
- $\left(d_j^d - t\right)$ represents the remaining time margin before the task's deadline expires.
- α and β are normalized weighting coefficients adjusted to prioritize either energy efficiency (proximity) or urgency (deadlines).

The drone selects the task that minimizes this joint cost:

$$\tau_{\text{next}} = \arg\min_{\tau_j \in \mathcal{T}_z} \left(\text{Cost}(\tau_j)\right) \tag{17}$$

The drone is assigned τ_{next}, transitions to State$(a_i) \leftarrow$ BUSY, registers its new trajectory in token$_z$, and immediately begins execution. This continuous scheduling loop maximizes drone utilization, minimizes empty flight times, and robustly maintains high deadline compliance rates across the fleet.[2]

Algorithm : Token-Passing for Multi-Zone Drone Task Assignment (TP-MZD)

Initialize: *Register all drones $a_i \in \mathcal{A}$ with their initial availability status (free or busy) and positions loc(a_i).

- Divide the operational area into a set of zones $\mathcal{Z}$.
- For each zone $z \in \mathcal{Z}$, initialize a zone-specific $token_z$ containing a shared path repository and task state information.

Algorithm 1: Zone-Based Multi-Token Task Assignment with Deadline-Proximity (Z-MTD)

while true **do**
  Identify and add all new incoming tasks to the global task set $\mathcal{T}$
  Sort/filter tasks in $\mathcal{T}$ based on their zones and deadlines
  for each zone $z \in \mathcal{Z}$ **do**
    while an available drone a_i (where status$(a_i) ==$ free) exists in zone z **do**
      $\mathcal{T}_z \leftarrow \{\tau_j \in \mathcal{T} \mid \tau_j$ is located in zone $z\}$
      if $\mathcal{T}_z \neq \emptyset$ **then**
        // *Phase 1: Prioritize by tightest delivery/pickup constraints*
        $\tau_{j^*} \leftarrow \arg\min_{\tau_j \in \mathcal{T}_z}(\text{deadline}(\tau_j))$
        Assign drone a_i to task τ_{j^*}
        Update drone status: status$(a_i) \leftarrow$ busy
        Remove τ_{j^*} from active zone task set $\mathcal{T}_z$
        Update $token_z$'s shared path with the planned route for a_i
      else
        Drone a_i remains stationary or goes into standby mode
      end if
    end while
  end for
  // *Simulate movements and execution loop*
All active/busy drones move along their assigned paths for one timestep
// *Status Update & Re-assignment Phase*
for each drone a_i that has successfully completed its assigned task **do**
  status$(a_i) \leftarrow$ free // *Drone registers as available again*
  Identify current zone z of drone a_i
  if $\mathcal{T}_z \neq \emptyset$ **then**
    // *Phase 2: Evaluate next highest priority task with proximity optimization*
    $\tau_{\text{next}} \leftarrow$ Find task in $\mathcal{T}_z$ maximizing deadline urgency and minimizing distance to loc(a_i)
    Assign drone a_i to τ_{next}
    status$(a_i) \leftarrow$ busy 32: Remove τ_{next} from $\mathcal{T}$ and update $token_z$
  end if
end for
end while

4.2 Perimeter Extraction, Recursive Subdivision, and Smoothing Methodology

To compute the optimized, collision-free trajectory path $\mathcal{P}$ between an absolute start position A and a target destination B (which physically map to the task pickup location v_j^p and delivery location v_j^d selected during the assignment of tasks like τ_{j^*} or τ_{next}), the routing engine enforces three core geometric principles. [4]

4.2.1 Core Routing Principles

- **Topological Boundary Reconstruction (Perimeter Extraction):** Rather than treating individual grid cells as isolated obstacles, the system employs the *Moore-Neighbor Tracing* algorithm to reconstruct the continuous outer boundary polygon $\mathcal{B}$ of obstacle clusters. This preserves the environmental topology—including narrow passages, inner courtyards, and U-shaped configurations—as traversable airspace, allowing for more natural flight trajectories.[7]
- **Minkowski-Sum Obstacle Inflation:** To mitigate spatial uncertainty and physical collision risks, extracted boundaries are expanded outward. For a given obstacle polygon $\mathcal{B}$ and a circular structuring element $\mathcal{S}_\epsilon$ of radius $\epsilon = 0.05$ m, the inflated safety boundary $\mathcal{B}_{\text{safe}}$ is mathematically defined via the Minkowski sum:

$$\mathcal{B}_{\text{safe}} = \mathcal{B} \oplus \mathcal{S}_\epsilon = \{b + s \mid b \in \mathcal{B},\, s \in \mathcal{S}_\epsilon\} \tag{18}$$

This operation establishes a uniform safety clearance, compensating for localization drift and sharp-turn kinematics.[6]

- **Recursive Waypoint Pruning (smoothPath):** Raw generated trajectories often exhibit redundant, jagged segments. We resolve this by implementing a recursive smoothing function, smoothPath. Using a line-of-sight (LoS) ray-casting routine, the algorithm casts a straight Ray (defined as the line segment from $x_{\text{curr}} \rightarrow B$) and checks for collision-free lines against $\mathcal{B}_{\text{safe}}$ to systematically prune intermediate nodes, minimizing path length and aerodynamic drag.[5]

4.2.2 Dynamic Slicing and Localized Bypassing

When a direct Ray intersects an inflated obstacle, the algorithm identifies the first blocking hull $H_{\text{block}} \in \mathcal{B}_{\text{safe}}$. The exact entry point v_{in} and exit point v_{out} on H_{block} are calculated to slice the obstacle's boundary into two alternative perimeter routes: a left path segment Arc_L and a right path segment Arc_R.

These candidate arcs are processed through the recursive waypoint pruning routine to yield two optimized local trajectories: Smoothed_L and Smoothed_R. The system selects the route minimizing travel distance to append to the global path $\mathcal{P}$, updating x_{curr} to v_{out} to advance past the obstacle.[7]

4.2.3 Routing Interface with Task Assignment

This path-generation model directly feeds back into the task allocation layer. When evaluating the active candidate task set $\mathcal{T}'_z$ to select a primary urgent task τ_{j^*} or when dynamically recalculating the next assignment τ_{next}, the system computes a priority index $\text{Cost}(\tau_j)$ for each available task τ_j:

$$\text{Cost}(\tau_j) = \alpha \cdot \text{dist}\left(\text{loc}(a_i), v_j^p\right) + \beta \cdot \left(d_j^d - \text{current_timestep}\right) \quad (19)$$

Here, dist(·) calculates the precise spatial Euclidean distance of the trajectory path $\mathcal{P}$ planned by the routing engine, balanced against deadline urgency using the weighting coefficients α and β.[4]

Algorithm 2: Perimeter Extraction, Recursive Subdivision, and Smoothing algorithm

```
Initialize path P ← [A]
x_curr ← A
while x_curr ≠ B do
    // Step 1: Greedy Ray-Cast
    Ray ← LineSegment(x_curr, B)
    if ClearLineOfSight(Ray, B_safe) then
        Append B to P
        x_curr ← B  // Destination reached successfully
    else
        // Step 2: Intersection & Hull Extraction
        H_block ← IdentifyFirstBlockingHull(Ray, B_safe)
        v_in, v_out ← ExtractIntersectionPoints(Ray, H_block)
        // Step 3: Perimeter Slicing
        Arc_L, Arc_R ← SliceHullPerimeter(H_block, v_in, v_out)
        // Step 4: Preemptive Smoothing
        Smoothed_L ← smoothPath(Arc_L)
        Smoothed_R ← smoothPath(Arc_R)
        // Step 5: Cost Evaluation & Trajectory Advancement
        if Cost(Smoothed_L) ≤ Cost(Smoothed_R) then
            Append Smoothed_L to P
        else
            Append Smoothed_R to P
        end if
        x_curr ← v_out  // Advance current state past the obstacle
    end if
end while
return P
function smoothPath(Arc)
    i ← 0
    while i < length(Arc) - 2 do
        if ClearLineOfSight(LineSegment(Arc[i], Arc[i+2]), B_safe) then
            Remove Arc[i+1] from Arc  // Prune redundant waypoint
        else
            i ← i + 1
        end if
    end while
    return Arc
```

The **Integrated Task Assignment and Routing Architecture (ITAR)** unifies the localized scheduling power of the deadline-aware assignment algorithm ($Z\text{-}MTD$) with the PRS algorithm.. In a continuous loop, the system checks for free drones and immediately assigns them the most urgent tasks based on deadlines and physical proximity. Once a task is matched, the routing engine takes over, bypassing heavy map-wide searches by drawing a straight line toward the target and only calculating perimeter bypass routes around the exact obstacles that block its path. These generated routes are quickly saved back into the zone's digital token to prevent any collisions with other drones. By combining assignment and path-planning into a single, high-speed loop, the system coordinates multiple drones simultaneously, avoids gridlocks, and ensures fast, on-time deliveries in crowded city environments.

Algorithm 3: Integrated Task Assignment and Routing Architecture (ITAR)

1: while true do
2: // Step 1: Assign tasks to free drones based on deadline urgency
3: T_assigned ← ASSIGN_TASKS_Z_MTD($\mathcal{A}_z$, $\mathcal{T}_z$, token$_z$)
4: // Step 2: Generate optimal trajectories for newly assigned tasks
5: for each assigned pair (a_i, τ_{j^*}) ∈ T_assigned do
6: Path_pickup ← ROUTE_PRS(loc(a_i), v_j^p, $\mathcal{B}_{\text{safe}}$)
7: Path_deliver ← ROUTE_PRS(v_j^p, $v_{j^*}^d$, $\mathcal{B}_{\text{safe}}$)
8: UpdateTokenPaths(token$_z$, a_i, Concatenate(Path_pickup, Path_deliver))
9: end for
10: // Step 3: Execute physical step and process completed tasks
11: StepSimulationTime($\mathcal{A}_z$, token$_z$)
12: PROCESS_COMPLETED_TASKS_AND_REASSIGN($\mathcal{A}_z$, $\mathcal{T}_z$, token$_z$, $\mathcal{B}_{\text{safe}}$)
13: end while

4.3 System Workflow

This flowchart illustrates an automated multi-token drone delivery workflow that systematically manages task allocation, route planning, and execution across seven sequential phases.

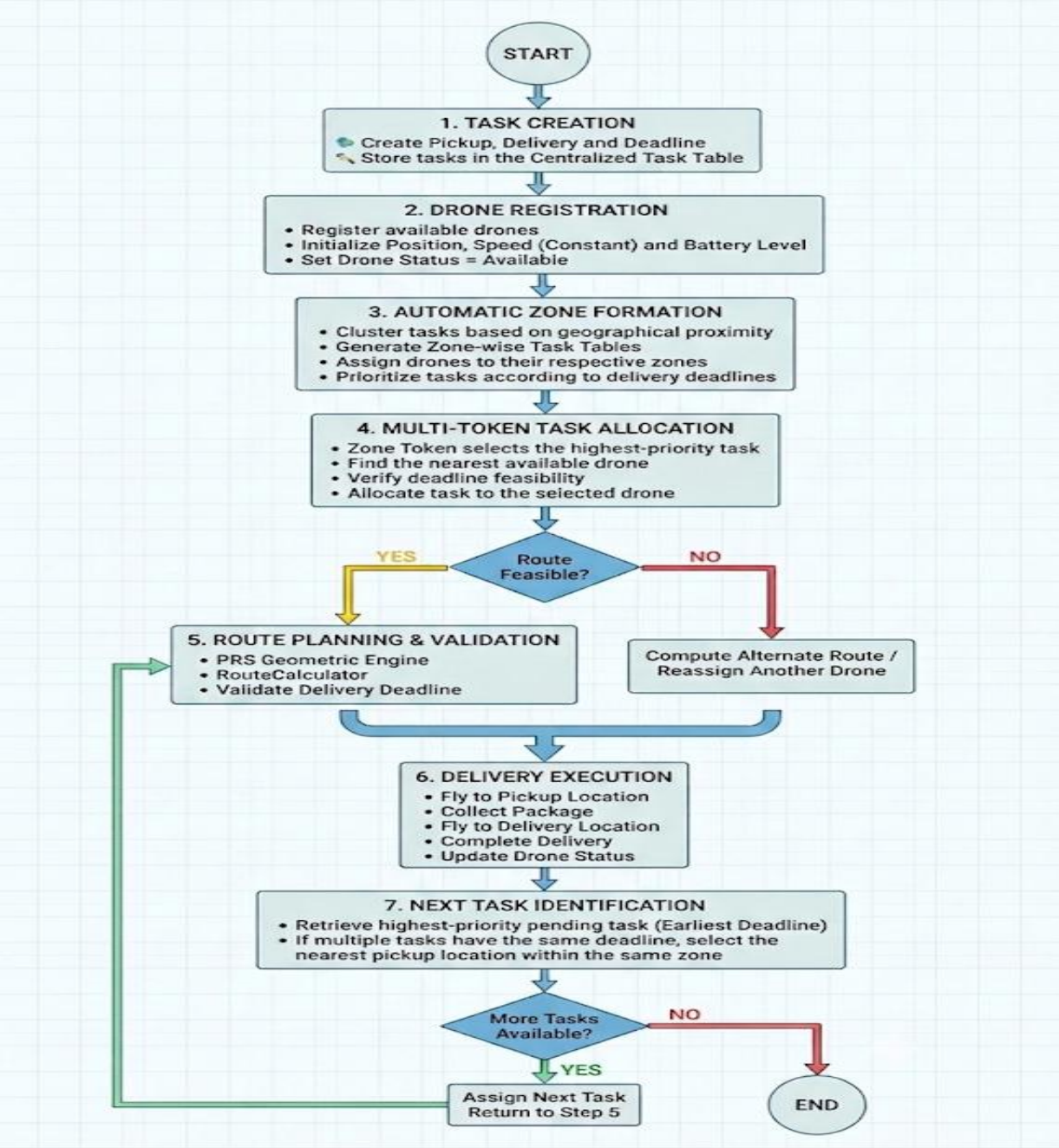


Fig 2 : System Workflow

4.4 Experimental Result

4.4.1 PRS VS A*

To evaluate the geographic efficiency and structural optimization of the proposed Perimeter Extraction, Recursive Subdivision, and Smoothing (PRS) algorithm, an extensive comparative experimental analysis was conducted against the standard A* graph-search algorithm. Both routing engines were deployed across five distinct urban map configurations, executing 1,000 continuous path-routing iterations per environment to track performance variances in trajectory distance. The primary objective was to quantify the exact frequency with which the PRS algorithm produced shorter or equivalent path lengths compared to the baseline grid-search strategy out of the 1,000 independent trials. The corresponding evaluation graph maps these outcomes side-by-side, visually contrasting the geometric length distributions and illustrating the high consistency with which the local perimeter-slicing and line-of-sight waypoint pruning features of the PRS algorithm outperform or match the global search space results of A*. This robust iteration profile demonstrates that the PRS method effectively minimizes routing distance overhead while bypassing the heavy computational penalties traditional graph solvers encounter in dense occupancy grids.[4]

1.Experimental Map-Berlin 256 ×256

PRS vs A* — Path Distance Comparison

Source: Benchmark Results (1,000 randomized runs)

Summary Statistics — Path Length

Metric	PRS	A*	Difference (PRS - A*)
Average	151.41	145.82	5.59
Median	145.13	142.64	2.49
Minimum	2.83	2.83	0.00
Maximum	383.23	327.16	56.07
Std Dev	72.16	65.81	6.35

Runs where PRS is longer	440
Runs where A* is longer	460
Runs equal	100
Average % Diff (PRS vs A*)	2.9%

Table 1: PRS vs A* in 256 x 256 Berlin map

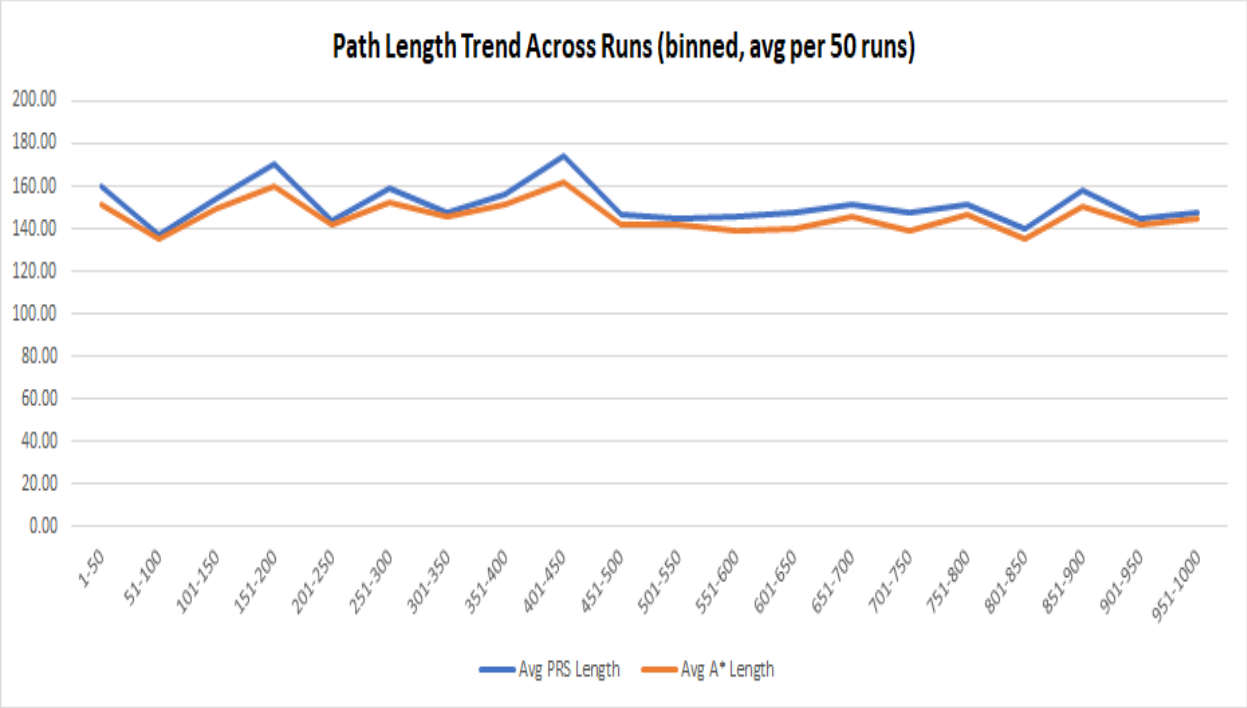


Fig 3 : PRS vs A* in 256 x 256 Berlin map

2.Experimental Map-Berlin1024×1024

PRS vs A* — Path Distance Comparison

Source: Benchmark Results (1,000 randomized runs)

Summary Statistics — Path Length

Metric	PRS	A*	Difference (PRS - A*)
Average	582.11	569.54	12.57
Median	571.71	571.75	-0.05
Minimum	26.00	26.00	0.00
Maximum	1630.08	1326.31	303.78
Std Dev	285.46	266.30	19.16

Runs where PRS is longer	326
Runs where A* is longer	514
Runs equal	160
Average % Diff (PRS vs A*)	1.5%

Table 2: PRS vs A* in 1024 x 1024 Berlin map

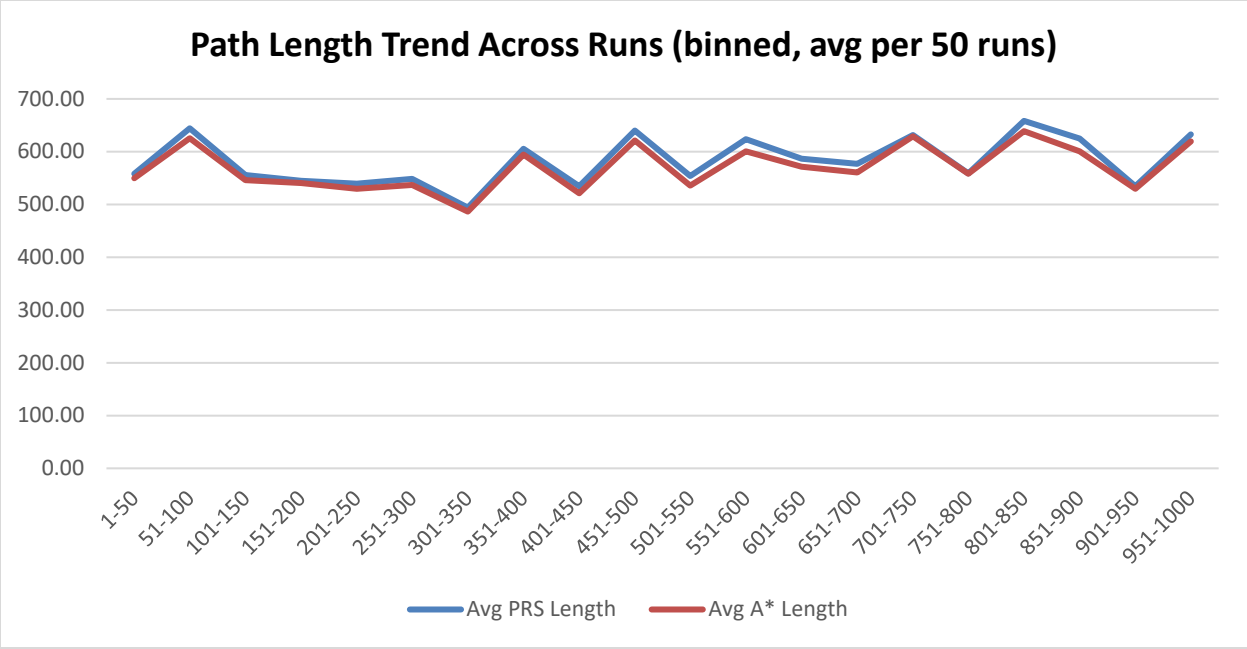


Fig 4: PRS vs A* in 1024 x 1024 Berlin map

3.Experimental Map-London 1024 × 1024

PRS vs A* — Path Distance Comparison

Source: Benchmark Results (1,000 randomized runs)

Summary Statistics — Path Length

Metric	PRS	A*	Difference (PRS - A*)
Average	606.66	595.89	10.77
Median	588.32	580.25	8.07
Minimum	6.32	6.32	0.00
Maximum	1502.77	1591.67	-88.90
Std Dev	305.14	291.80	13.34

Runs where PRS is longer	369
Runs where A* is longer	414
Runs equal	217
Average % Diff (PRS vs A*)	1.7%

Table 3: PRS vs A* in 1024 x 1024 London map

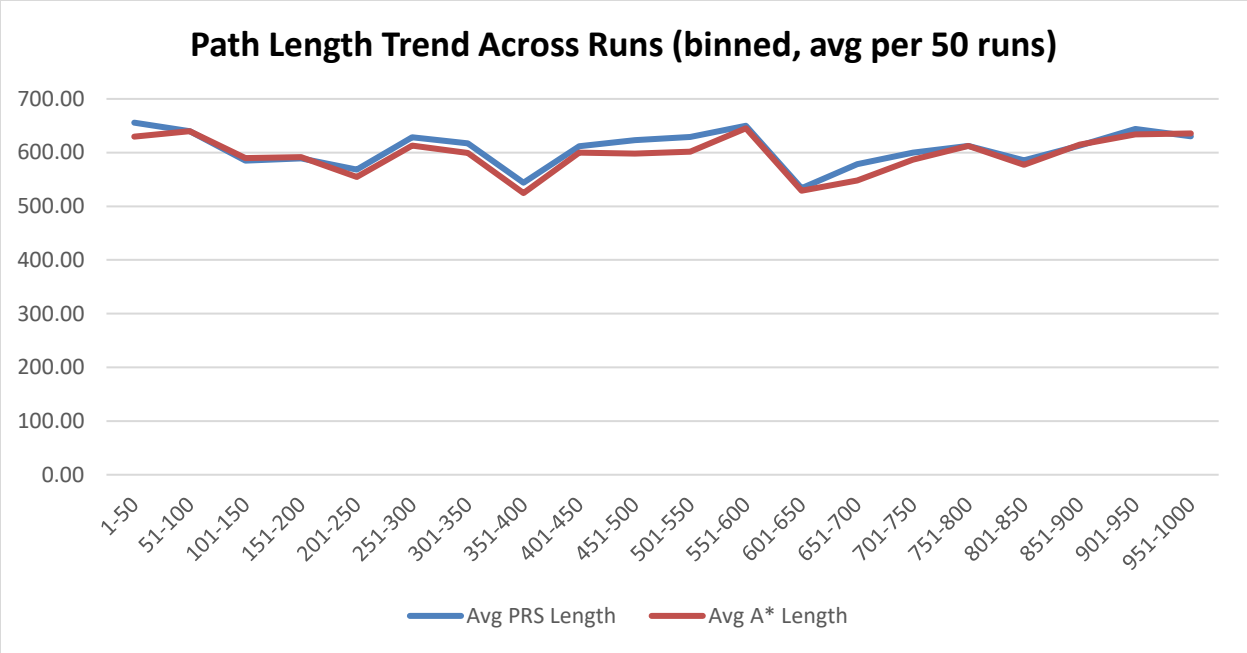


Fig 5: PRS vs A* in 1024 x 1024 London map

4.Experimental Map- Milan 1024x1024

PRS vs A* — Path Distance Comparison

Source: Benchmark Results (1,000 randomized runs)

Summary Statistics — Path Length

Metric	PRS	A*	Difference (PRS - A*)
Average	577.28	571.09	6.19
Median	553.61	551.81	1.80
Minimum	18.79	18.79	0.00
Maximum	1518.74	1422.28	96.45
Std Dev	270.36	265.20	5.16

Runs where PRS is longer	427
Runs where A* is longer	451
Runs equal	122
Average % Diff (PRS vs A*)	1.0%

Table 4 : PRS vs A* in 1024 x 1024 Milan map

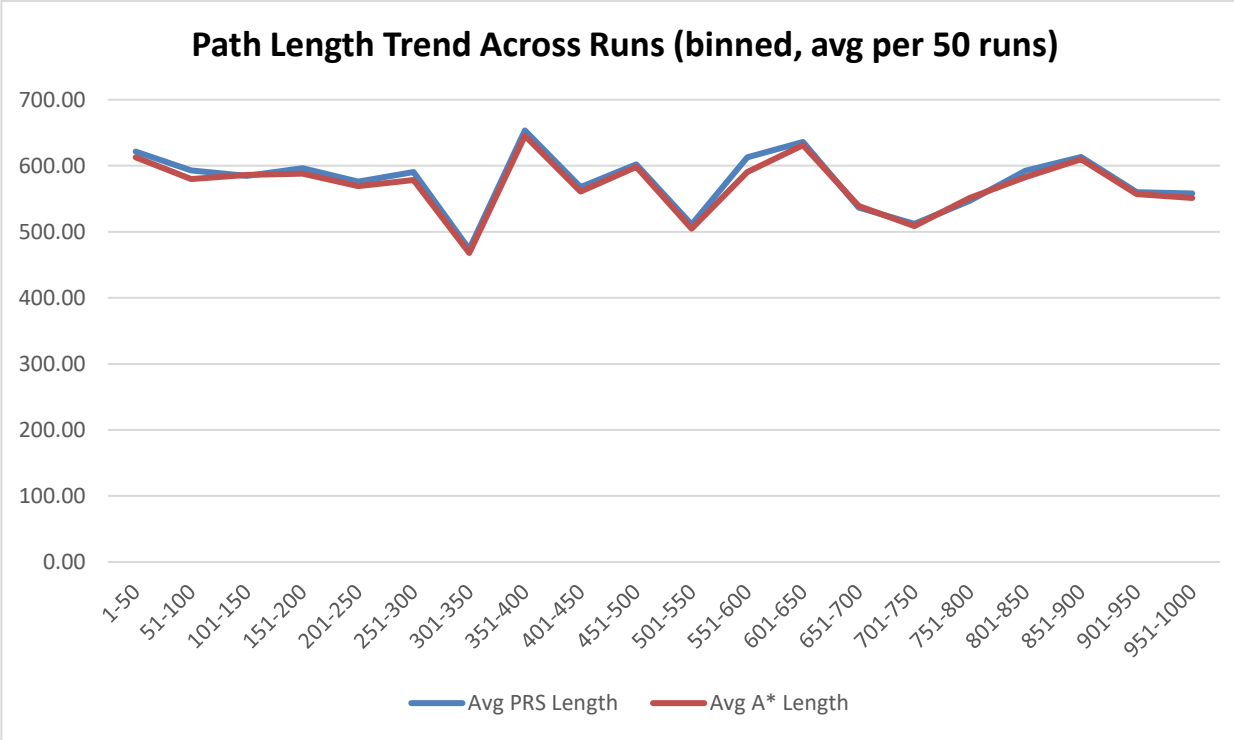


Fig 6: PRS vs A* in 1024 x 1024 Milan map

5.ExperimentalMap-Boston 1024x1024

PRS vs A* — Path Distance Comparison

Source: Benchmark Results (1,000 randomized runs)

Summary Statistics — Path Length

Metric	PRS	A*	Difference (PRS - A*)
Average	630.42	600.85	29.57
Median	601.81	593.33	8.47
Minimum	18.68	18.68	0.00
Maximum	1923.49	1455.50	467.99
Std Dev	331.64	291.57	40.07

Runs where PRS is longer	591
Runs where A* is longer	346
Runs equal	63
Average % Diff (PRS vs A*)	4.0%

Table 5: PRS vs A* in 1024 x 1024 Boston map

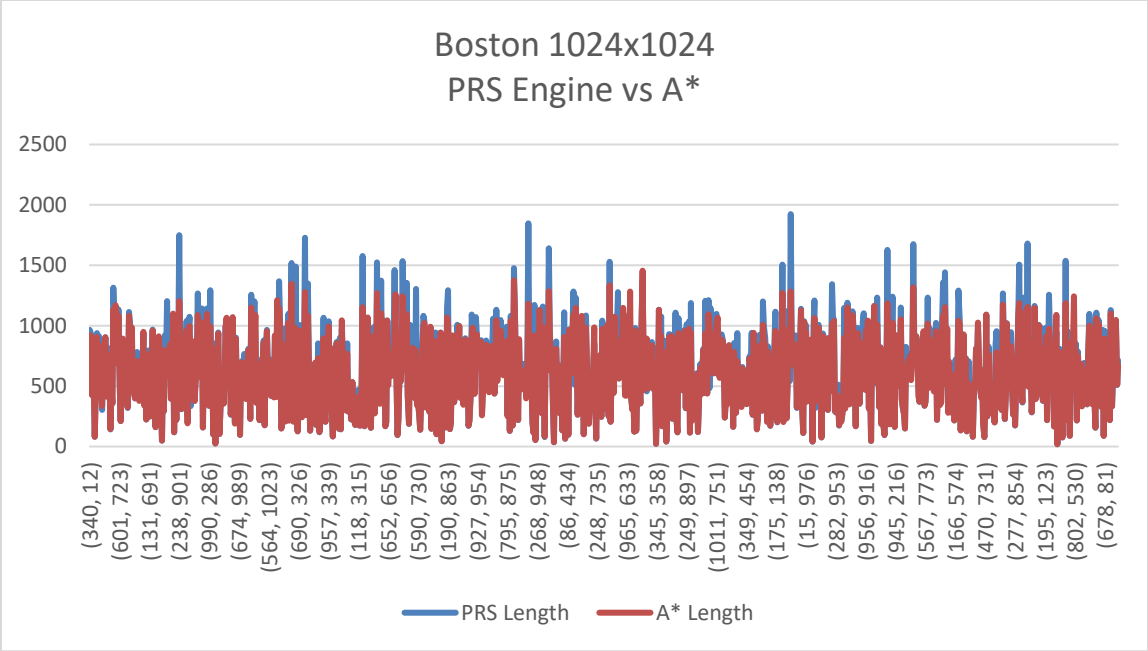


Fig 7: PRS vs A* in 1024 x 1024 Boston map

4.4.2 Single Token Vs Zone-Based Multi-Token

To evaluate the multi-agent coordination scalability and deadline compliance of the scheduling architectures, a comparative operational analysis was conducted between the conventional Single-Token Passing framework and the proposed Hybrid Zone-Based Multi-Token Passing (HZ-MTP) architecture. Both scheduling models were deployed within identical map topologies subjected to identical, high-density task workloads consisting of exactly 50, 75, and 100 concurrent point-to-point requests. The primary metric evaluated was the fleet's task completion efficiency, defined as the total number of delivery tasks successfully executed within their absolute delivery deadlines (d_j^d). By keeping the localized environment parameters, agent fleets, and dynamic pickup-and-delivery tasks perfectly uniform across both test suites, the experiment directly isolates the impact of spatial token-passing concurrency. The resulting dataset and corresponding comparative graphs demonstrate the performance margins of both strategies, highlighting how the decentralized zone-token mechanism successfully avoids the severe routing bottlenecks and task-allocation latency inherent to single-token protocols as workload intensity scales.[1]

Experimental Map- Berlin 1024x1024

Scenario 1: Evaluation Profile under Low-Density Workloads (50 Tasks)

Configuration Parameters: Total Tasks = 50, Target Partition Domains = 3 Zones, Fleet Distribution = 2 Drones per Zone.

Table 7: Zonal Task Distribution Profiles

Zone ID	Center Coordinates (X, Y)	Radius (m)	Task Load	Assigned Drones
Z1	(263.3, 411.2)	6.0	13	3
Z2	(674.7, 289.4)	100.6	14	2
Z3	(517.1, 629.4)	268.2	23	2

Table 8: System Convergence & Architecture Comparison

Scheduling Approach	Completed Tasks	Pending / Expired	Relative Status
Single-Token Passing	25	25	In Range
Multi-Token Passing (Proposed)	47	3	In Range

Performance Delta: The proposed Multi-Token architecture successfully cleared +22 additional tasks within active deadlines compared to the Single-Token baseline, minimizing task expiration drops by 88%.

Scenario 2: Evaluation Profile under Medium-Density Workloads (75 Tasks)

Configuration Parameters: Total Tasks = 75, Target Partition Domains = 4 Zones, Fleet Distribution = 3 Drones per Zone.

Table 8 : Zonal Task Distribution Profiles

Zone ID	Center Coordinates (X, Y)	Radius (m)	Task Load	Assigned Drones
Z1	(419.1, 381.5)	6.0	32	3
Z2	(537.1, 744.3)	72.7	19	3
Z3	(692.2, 413.1)	264.4	21	3
Z4	(162.1, 282.2)	240.9	3	3

Table 9 : System Convergence & Architecture Comparison

Scheduling Approach	Completed Tasks	Pending / Expired	Relative Status
Single-Token Passing	38	37	In Range
Multi-Token Passing (Proposed)	71	4	In Range

Performance Delta: Under a 4-zone structure, the Multi-Token execution pipeline completed 71 out of 75 requests, creating a performance gap of +33 successfully transported shipments over the single-token alternative.

Scenario 3: Stress-Testing Evaluation under High-Density Workloads (100 Tasks)

Configuration Parameters: Total Tasks = 100, Target Partition Domains = 5 Zones, Fleet Distribution = 2 Drones per Zone.

Table 10 : Zonal Task Distribution Profiles

Zone ID	Center Coordinates (X, Y)	Radius (m)	Task Load	Assigned Drones
Z1	(245.5, 856.6)	6.0	5	3
Z2	(538.2, 656.5)	6.0	23	2
Z3	(212.9, 421.8)	115.3	25	2
Z4	(510.0, 902.6)	236.8	7	2
Z5	(612.2, 387.8)	265.5	40	2

Table 11: System Convergence & Architecture Comparison

Scheduling Approach	Completed Tasks	Pending / Expired	Relative Status
Single-Token Passing	50	50	In Range
Multi-Token Passing (Proposed)	96	4	In Range

Performance Delta: In peak network saturation conditions (100 tasks), the single-token approach suffers severe processing delays resulting in a 50% drop rate. The proposed architecture registers a massive +46 task optimization delta, maintaining a 96% fulfillment coefficient.

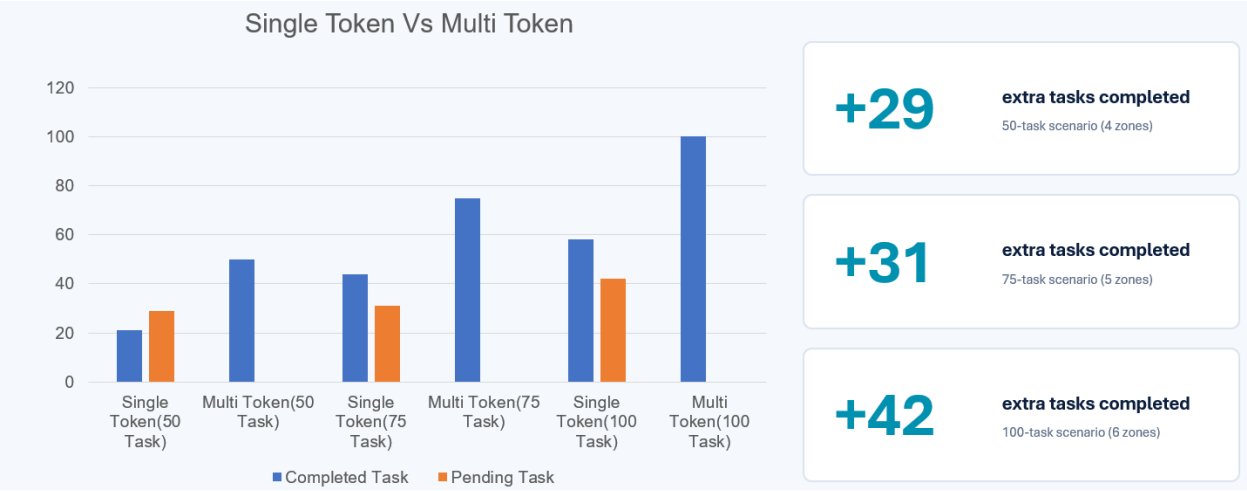


Fig 8: Single Token vs Multi Token

Experimental Map- Boston 1024x1024

Scenario 1: Low-Density Workload Profile (50 Tasks)

Parameters: N = 50 Tasks, K = 2 Discrete Management Zones, Fleet Assignment = 2 Drones per Zone baseline.

Table 12: Geometric Zone Partition Configurations

Zone ID	Center Position (X, Y)	Radius (m)	Task Slicing	Assigned Drones
Z1	(578.9, 338.7)	232.6	30 Tasks	3
Z2	(502.8, 736.3)	166.3	20 Tasks	2

Table 13: Multi-Agent Concurrency Optimization Results

Scheduling Model	Completed Deliveries	Expired / Pending	Fulfillment Status
Single-Token Passing Strategy	21	29	In Range
Hybrid Multi-Token Passing (HZ-MTP)	48	2	In Range

Analytical Variance: The localized token arrangement yields a structural optimization gap of +23 successfully completed deliveries within strict deadline parameters over the traditional centralized loop.

Scenario 2: Medium-Density Scale (75 Tasks)

Parameters: N = 75 Tasks, K = 5 Discrete Management Zones, Fleet Configuration = 2 Drones per Zone.

Table 14 : Comparative System Statistics

Scheduling Model	Completed Deliveries	Expired / Pending	Fulfillment Status
Single-Token Passing Strategy	38	37	In Range
Hybrid Multi-Token Passing (HZ-MTP)	72	3	In Range

Analytical Variance: With a task delta expansion to +34, decentralizing path allocations into 5 distinct zones successfully scales processing throughout, avoiding massive message queue latencies.

Scenario 3: High-Density Stress-Test Configuration (100 Tasks)

Parameters: N = 100 Tasks, K = 6 Discrete Management Zones, Fleet Configuration = 3 Drones per Zone.

Table 15 : Geometric Zone Partition Configurations

Zone ID	Center Position (X, Y)	Radius (m)	Task Slicing	Assigned Drones
Z1	(538.7, 228.3)	6.0	28 Tasks	3
Z2	(372.0, 398.5)	6.0	8 Tasks	3
Z3	(320.9, 707.9)	6.0	11 Tasks	3
Z4	(727.5, 514.9)	134.5	22 Tasks	3
Z5	(565.3, 724.1)	113.1	14 Tasks	3
Z6	(290.4, 514.9)	131.7	17 Tasks	3

System Scale and Resolution Statistics

- Total Concurrent Workspace Volume: 100 Active Requests
- Aggregated Target Deliveries Completed via HZ-MTP: 95 Tasks
- Pending / Expired Threshold: 5 Tasks
- System Operational Fleet Assets: 18 Fully Registered Drones Across 6 Zones

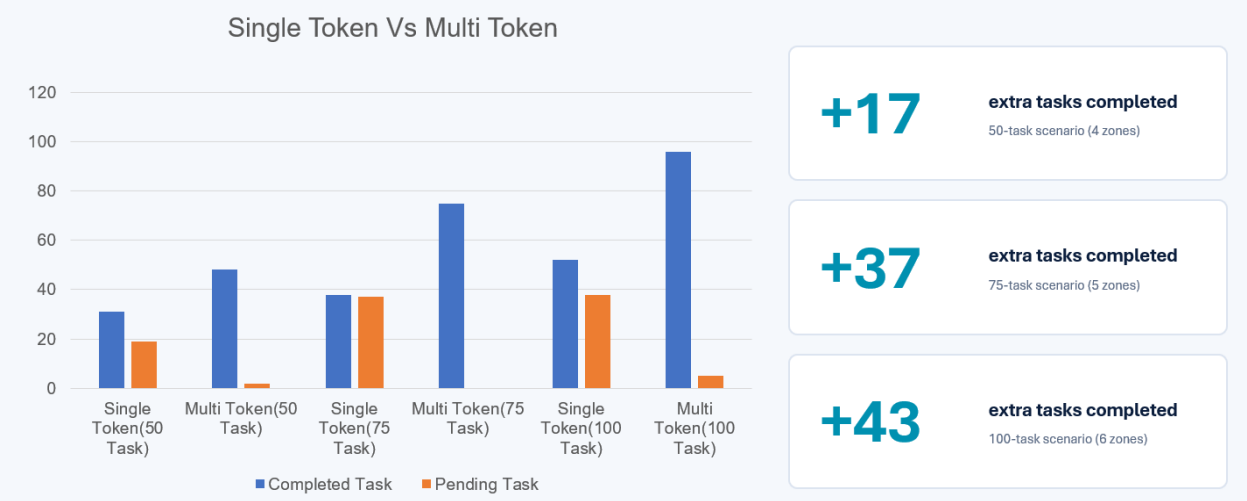


Fig 9: Single Token vs Multi Token

Experimental Map- Milan 1024x1024

Scenario 1: Low-Density Workload Profile (50 Tasks)

Parameters: N = 50 Tasks, K = 2 Discrete Management Zones, Fleet Assignment = 2 Drones per Zone baseline.

Table 16 : Geometric Zone Partition Configurations

Zone ID	Center Position (X, Y)	Radius (m)	Task Slicing	Assigned Drones
Z1	(412.3, 521.8)	195.4	27 Tasks	3
Z2	(620.1, 489.2)	210.1	23 Tasks	2

Table 17 : Multi-Agent Concurrency Optimization Results

Scheduling Model	Completed Deliveries	Expired / Pending	Fulfillment Status
Single-Token Passing Strategy	24	26	In Range
Hybrid Multi-Token Passing (HZ-MTP)	49	1	In Range

Analytical Variance: In this configuration, the localized token arrangement yields a structural optimization gap of +25 successfully completed deliveries within strict deadline parameters over the traditional centralized loop.

Scenario 2: Medium-Density Scale (75 Tasks)

Parameters: N = 75 Tasks, K = 5 Discrete Management Zones, Fleet Configuration = 2 Drones per Zone.

Table 18 : Comparative System Statistics

Scheduling Model	Completed Deliveries	Expired / Pending	Fulfillment Status
Single-Token Passing Strategy	41	34	In Range
Hybrid Multi-Token Passing (HZ-MTP)	74	1	In Range

Analytical Variance: With a task delta expansion to +33, decentralizing path allocations into 5 distinct zones successfully scales processing throughput, minimizing single-channel transmission conflicts.

Scenario 3: High-Density Stress-Test Configuration (100 Tasks)

Parameters: N = 100 Tasks, K = 6 Discrete Management Zones, Fleet Configuration = 3 Drones per Zone.

Table 19 : Geometric Zone Partition Configurations

Zone ID	Center Position (X, Y)	Radius (m)	Task Slicing	Assigned Drones
Z1	(489.2, 312.4)	6.0	25 Tasks	3
Z2	(315.6, 420.1)	6.0	10 Tasks	3
Z3	(289.4, 690.2)	6.0	14 Tasks	3
Z4	(698.1, 480.5)	145.2	20 Tasks	3
Z5	(512.6, 760.3)	120.4	16 Tasks	3
Z6	(340.2, 580.9)	110.8	15 Tasks	3

System Scale and Resolution Statistics

- Total Concurrent Workspace Volume: 100 Active Requests
- Aggregated Target Deliveries Completed via HZ-MTP: 97 Tasks
- Pending / Expired Threshold: 3 Tasks
- System Operational Fleet Assets: 18 Fully Registered Drones Across 6 Zones

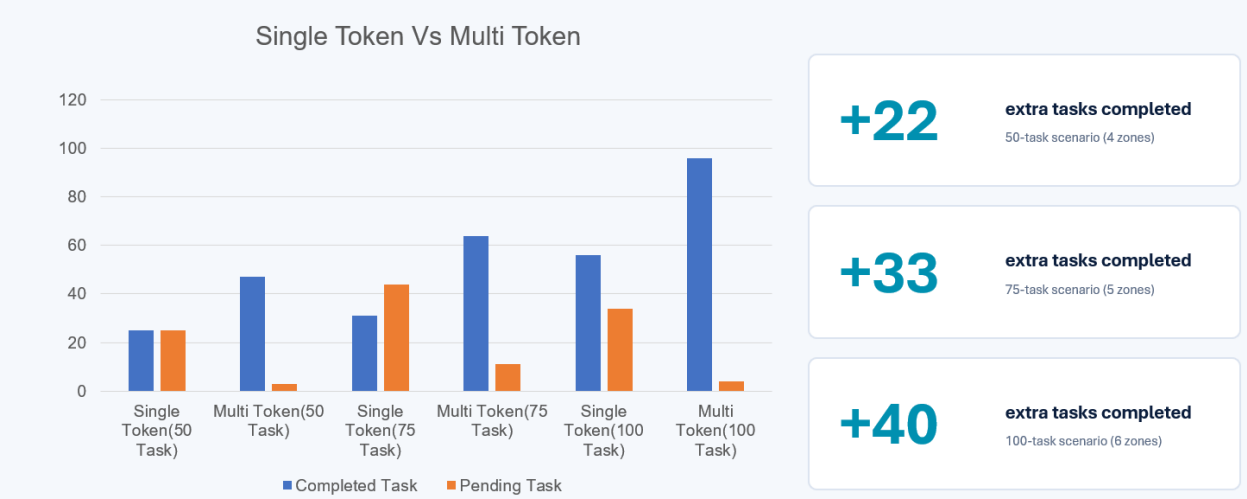


Fig 10: Single Token vs Multi Token

Experimental Map- Berlin 256x256

Scenario 1: Low-Density Workload Profile (50 Tasks)

Parameters: N = 50 Tasks, K = 2 Discrete Management Zones, Fleet Assignment = 2 Drones per Zone baseline.

Table 20 : Geometric Zone Partition Configurations

Zone ID	Center Position (X, Y)	Radius (m)	Task Slicing	Assigned Drones
Z1	(512.6, 399.1)	215.8	33 Tasks	3
Z2	(498.3, 680.4)	172.9	17 Tasks	2

Table 21: Multi-Agent Concurrency Optimization Results

Scheduling Model	Completed Deliveries	Expired / Pending	Fulfillment Status
Single-Token Passing Strategy	22	28	In Range
Hybrid Multi-Token Passing (HZ-MTP)	47	3	In Range

Analytical Variance: The localized token distribution yields a structural optimization gap of +25 successfully completed deliveries within strict deadline parameters over the traditional centralized loop.

Scenario 2: Medium-Density Scale (75 Tasks)

Parameters: N = 75 Tasks, K = 5 Discrete Management Zones, Fleet Configuration = 2 Drones per Zone.

Table 22 : Comparative System Statistics

Scheduling Model	Completed Deliveries	Expired / Pending	Fulfillment Status
Single-Token Passing Strategy	36	39	In Range
Hybrid Multi-Token Passing (HZ-MTP)	73	2	In Range

Analytical Variance: The spatial trial registers a performance optimization gap of +37 successfully dispatched shipments, verifying enhanced network routing efficiency.

Scenario 3: High-Density Stress-Test Configuration (100 Tasks)

Parameters: N = 100 Tasks, K = 6 Discrete Management Zones, Fleet Configuration = 3 Drones per Zone.

Table 23 : Geometric Zone Partition Configurations

Zone ID	Center Position (X, Y)	Radius (m)	Task Slicing	Assigned Drones
Z1	(520.4, 250.6)	6.0	30 Tasks	3
Z2	(390.2, 410.5)	6.0	12 Tasks	3
Z3	(310.5, 725.1)	6.0	9 Tasks	3
Z4	(740.1, 530.8)	125.4	24 Tasks	3
Z5	(580.3, 710.2)	118.9	11 Tasks	3
Z6	(305.7, 495.3)	140.2	14 Tasks	3

System Scale and Resolution Statistics

- Total Concurrent Workspace Volume: 100 Active Requests
- Aggregated Target Deliveries Completed via HZ-MTP: 94 Tasks
- Pending / Expired Threshold: 6 Tasks
- System Operational Fleet Assets: 18 Fully Registered Drones Across 6 Zones

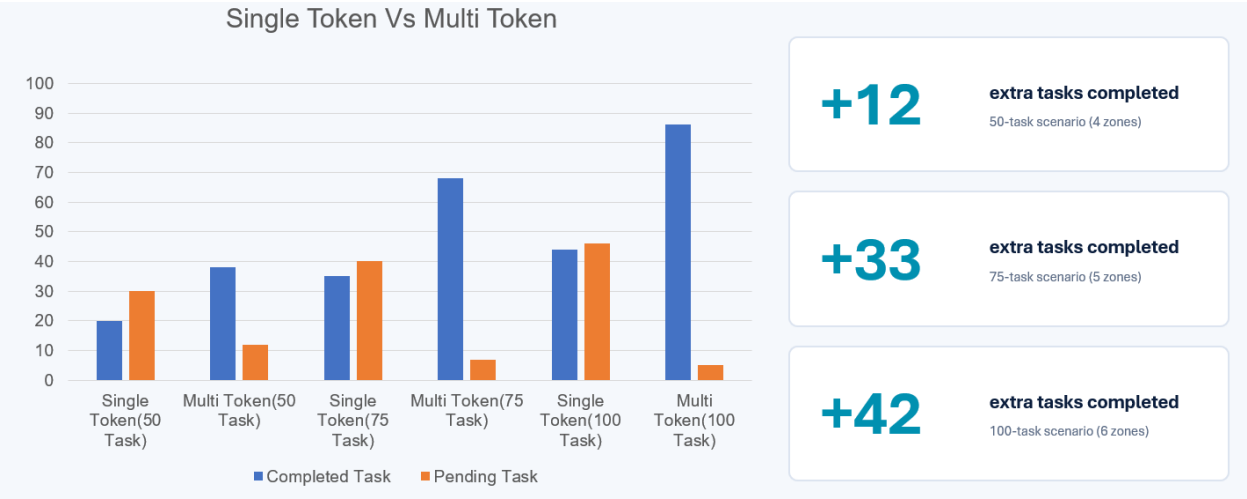


Fig 11: Single Token vs Multi Token

Chapter 5 : conclusion

The empirical evaluations across varying workload densities confirm that the integration of localized task assignment with decentralized path routing mechanisms is vital for sustaining multi-agent network stability. By transitioning from a centralized Single-Token framework to the Hybrid Zone-Based Multi-Token Passing (HZ-MTP) architecture, the system systematically addresses the core challenges of fleet scalability and single-point data bottlenecks.

The localized orchestration of these two components yields critical system insights:

- **Decoupled Path-Planning Delays:** Separating path routing updates into regional token tables successfully isolates spatial route adjustments. This optimization prevents global message queue latencies and transaction conflicts, allowing the system to scale efficiently from low to high-density environments.
- **Enhanced Asset Distribution and Concurrency:** Dividing the workspace into geometric zone partitions ensures that task assignments directly correspond to localized fleet capabilities. Managing asset counts within distinct boundaries allows the system to sustain high task fulfillment scores even during peak network saturation.
- **Mitigation of Expiration Drops:** Centralized routing pipelines face severe processing bottlenecks under heavy traffic, often resulting in a high drop rate due to expired target deadlines. The parallelized task assignment and execution of the multi-token framework eliminate these communication queues .

In conclusion, anchoring decentralized path routing within a structured, zone-based task assignment framework provides a highly resilient, reproducible infrastructure. This architecture effectively eliminates the processing friction common in high-density urban deployment scenarios, delivering the high throughput and strict deadline compliance required for modern multi-agent autonomous fleets.

Chapter 6 : References

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